MINI PROJECT

Detecting Fraud in Blockchain Transactions Using Machine Learning

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Problem Statement

Anonymity & Complexity

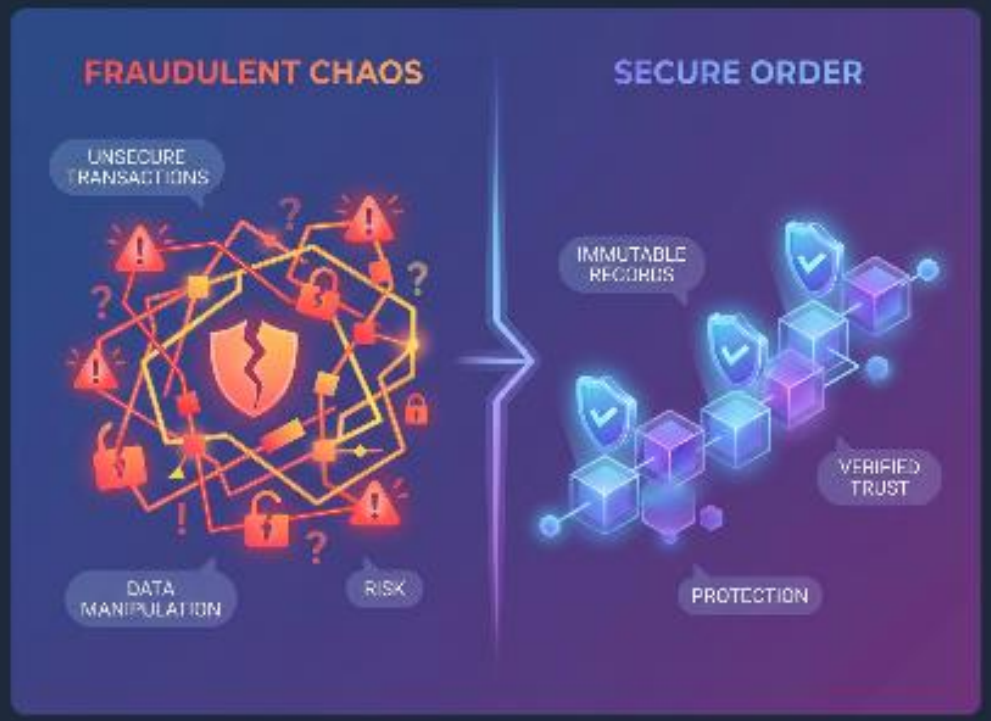
Blockchain transactions are decentralized and pseudonymous, obscuring user identities.

Hidden Patterns

Traditional rule-based methods fail to identify complex, hidden patterns in large-scale data.

Detection Difficulty

Manual review is impossible due to the sheer volume and velocity of transactions.



Visualizing the chaos of fraudulent patterns in a network

CORE PROBLEM

To identify and classify fraudulent blockchain transactions efficiently using advanced machine learning techniques.

Objectives

Analyze Patterns

To analyze decentralized blockchain transaction logs to identify hidden structural patterns and anomalies.

Classify Transactions

To classify transactions as fraudulent or non-fraudulent using supervised learning techniques.

Build ML Models

To build and train robust machine learning models capable of detecting fraud in large-scale datasets.

Evaluate Metrics

To evaluate model performance using key metrics like precision, recall, F1-score, and accuracy.

Improve Accuracy

To improve detection accuracy and significantly reduce false positives in transaction classification.

Literature Survey

Traditional ML Models

Logistic Regression, Random Forest, and SVMs used for extracting patterns from tabular transaction features.

Deep Learning Methods

LSTMs and CNNs utilized to detect sequential patterns and anomalies in large-scale time-series data.

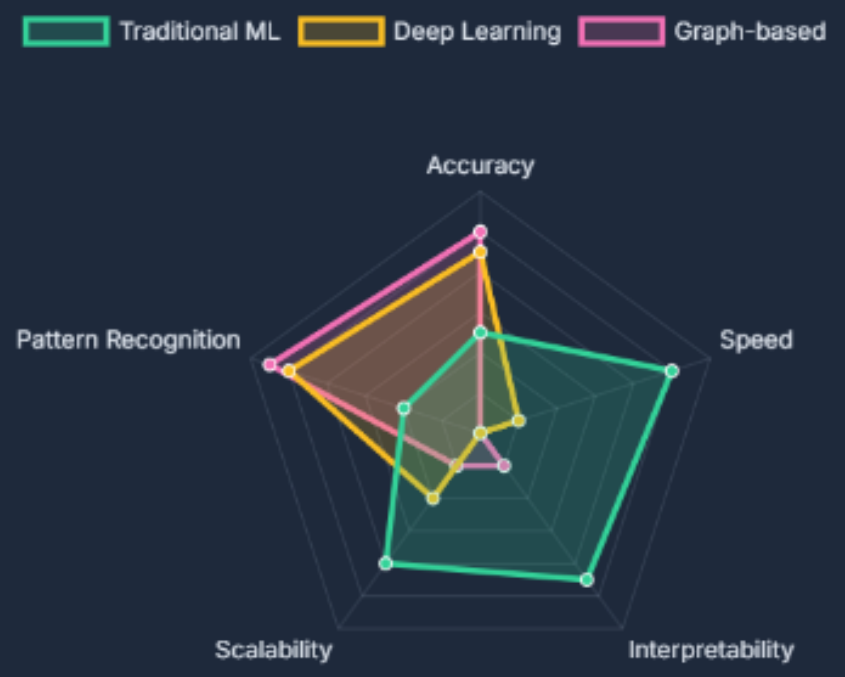
Graph-based Analysis

Network analysis techniques (GCNs) leverage the topology of blockchain transaction graphs to find fraud rings.

KEY CHALLENGES

- **Data Imbalance:** Fraud cases are extremely rare (<1%).
- **Label Scarcity:** Few verified fraudulent datasets exist.
- **High Cost:** Computational intensity of graph processing.

Methodology Comparison



Proposed System Overview

SYSTEM WORKFLOW

Input
Blockchain Transactions

→

Preprocessing
Cleaning & Normalization

→

Feature Eng.
Extracting Patterns

→

Training
ML Algorithms

→

Output
Fraud / Not Fraud

SYSTEM ADVANTAGES

Faster Detection
Real-time analysis enables immediate identification of suspicious activities.

Better Accuracy
Machine learning models reduce false positives compared to traditional rules.

Scalable System
Architecture designed to handle increasing transaction volumes efficiently.

Dataset Details

Data Source

Public blockchain ledgers (e.g., Ethereum/Bitcoin) accessed via Kaggle repositories or APIs.

Key Features

Transaction amount, Sender/Receiver addresses, Timestamp, Gas price, and Block height.

Preprocessing Pipeline

Imputing missing values, Min-Max Normalization of amounts, and Label Encoding.

TARGET VARIABLE

Binary classification output to train the model:
0 (Non-Fraudulent) vs 1 (Fraudulent)

transaction_data_sample.csv

```
// Feature Structure
{
  "tx_hash": "0x4a2...9d1",
  "timestamp": 1678892451,
  "from_addr": "0x8b1...2c4",
  "to_addr": "0x9c2...5e1",
  "amount_eth": 12.5042,
  "gas_price": 0.000032,
  "is_contract": false,
  "tx_count": 42,
  "label": 0 // Non-Fraud
}
```

Dataset size: ~500k records

JSON representation of a single transaction record

System Architecture

COMPLETE WORKFLOW

1. Data Collection
Blockchain Ledger

→

2. Preprocessing
Cleaning & Scaling

→

3. Feature Eng.
Pattern Extraction

4. Model Training
ML Algorithms

←

5. Prediction
Classify Transaction

←

6. Evaluation
Performance Metrics

Workflow Diagram

Tools & Technologies

Programming Language

Python: The primary language used for its robust data science ecosystem and ease of implementation.

Data Processing

Pandas & NumPy: Used for high-performance data manipulation, cleaning, and numerical analysis.

Machine Learning

Scikit-learn: Provides efficient tools for predictive data analysis and fraud classification models.

Data Visualization

Matplotlib & Seaborn: For creating static, animated, and interactive visualizations of transaction data.

Development IDE

Jupyter & VS Code: Integrated development environments for coding, testing, and debugging models.

Sample Input and Output

Input Features

The model processes raw transaction data including:

- Sender & Receiver Address
- Transaction Amount (Value)
- Timestamp & Gas Fee

Model Output

The system classifies each transaction into categories:

Fraudulent | Non-Fraudulent

LIVE PREDICTION LOG

Transaction Hash	Value (ETH)	Time	Prediction
0x7a2...9b1	50.00	10:42:01	FRAUD
0xb4c...2d9	2.45	10:42:05	CLEAN
0x1f8...3e4	1200.00	10:42:12	FRAUD
0x9d2...8a5	0.05	10:42:18	CLEAN
0xe3a...4b7	15.20	10:42:25	CLEAN

References

Research Papers

[1] Monamo, P., et al. (2016)
"Unsupervised Learning for Anomaly Detection in Bitcoin Network." *IEEE Symposium on Security and Privacy.*

[2] Chen, W., et al. (2018)
"Detecting Ponzi Schemes on Ethereum: Towards Healthier Blockchain Technology." *WWW '18 Proceedings.*

[3] Weber, M., et al. (2019)
"Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks." *arXiv preprint.*

[4] Farrugia, S., et al. (2020)
"Detection of Illicit Activity on the Bitcoin Blockchain." *Springer International.*

Datasets

Elliptic Data Set
A dataset of cryptographic currencies mapping Bitcoin transactions to real entities. Available on Kaggle.

Ethereum Fraud Detection Dataset
Collection of known fraudulent Ethereum addresses and transaction histories for supervised learning.

BitcoinHeist Ransomware
UCI Machine Learning Repository dataset containing labeled ransomware payments from 2009-2018.

Tools & Docs

Scikit-learn Documentation
Official user guide for ML algorithms (Random Forest, SVM, Isolation Forest).

Pandas & NumPy
Reference manuals for data manipulation and numerical analysis in Python.

Imbalanced-learn
Documentation for handling dataset imbalance (SMOTE, NearMiss techniques).

Web3.py
Library for interacting with Ethereum blockchain nodes.